

Function:

$$F(A, B, C, D) = \Sigma(1, 3, 7, 11, 15) + d(0, 2, 5)$$

	A	B	C	D	F	
0	0	0	0	0	X	X
1	0	0	0	1	1	
2	0	0	1	0	X	X
3	0	0	1	1	1	
4	0	1	0	0	0	X
5	0	1	0	1	X	
6	0	1	1	0	0	D
7	0	1	1	1	1	
8	1	0	0	0	0	0
9	1	0	0	1	0	
10	1	0	1	0	0	D
11	1	0	1	1	1	
12	1	1	0	0	0	0
13	1	1	0	1	0	
14	1	1	1	0	0	D
15	1	1	1	1	1	

A	B	C	F
0	0	0	X
0	0	1	X
0	1	0	X
0	1	1	D
1	0	0	0
1	0	1	D
1	1	0	0
1	1	1	D

BC	00	01	11	10
A	X ₀	X ₁	D ₃	X ₂
1	0 ₄	D ₅	D ₇	0 ₆

= $CD + \bar{A}D$

Step 2: Treat 1 as Don't Care

BC	00	01	11	10
A	X ₀	X ₁	D ₃	D ₂
1	0 ₄	D ₅	D ₇	0 ₆

= $CD + \bar{A}D$

$$F(A, B, C, D) = \sum(0, 2, 5, 8, 13, 15) + d(1, 10)$$

	A	B	C	D	F
0	0	0	0	0	1
1	0	0	0	1	X
2	0	0	1	0	1
3	0	0	1	1	0
4	0	1	0	0	0
5	0	1	0	1	1
6	0	1	1	0	0
7	0	1	1	1	0
8	1	0	0	0	1
9	1	0	0	1	0
10	1	0	1	0	X
11	1	0	1	1	0
12	1	1	0	0	0
13	1	1	0	1	1
14	1	1	1	0	0
15	1	1	1	1	1

A	B	C	F
0	0	0	X
0	0	1	D'
0	1	0	D
0	1	1	0
1	0	0	D'
1	0	1	X
1	1	0	D
1	1	1	D

BC	00	01	11	10
A				
0	0	1	3	2
1	4	5	7	6

Step 1: treat D as 0

BC	00	01	11	10
A				
0	1	0	0	0
1	0	0	0	0

$$= A'B'C'$$

Step 2: D

BC	00	01	11	10
A				
0	X	X	0	0
1	X	0	0	0

$$= A'B'C' + BC'D + ABD + A'B'D' + B'C'D'$$

$$BC'D + ABD$$

Step 3: D'

BC	00	01	11	10
A				
0	X	D'	0	X
1	D'	0	X	X

$$= A'B'D' + B'C'D'$$

